



Figure 6: Designer screen

know, is the cloud based database from Google. We will use it to store the user's data on the cloud.

Table 1 shows the components that are required for this application. We will drag them on to the designer from the left-hand side palette.

1. Drag and drop the components mentioned in Table 1 to the viewer.
2. Visible components will be visible to you, while the non-visible components will be located beneath the viewer, tagged as 'Non-visible'.
3. Labels along with their respective text boxes need to be put within the horizontal arrangement so as to keep them aligned horizontally.
4. If you have dragged and placed everything, the layout will look something like what's shown in Figure 6.
5. Make the necessary property changes like we did when changing the text property for the label and button components.
6. Renaming the components helps to identify them in the block editor.
7. Your graphical user interface is

Table 1

Component's name	Purpose	Location
Label	To display a label	Palette-->User Interface-->Label
Button	To trigger events	Palette-->User Interface-->Button
Horizontal arrangement	To arrange the child components	Palette-->Layout-->Horizontal Arrangement
Notifier	To display on-screen information	Palette-->User Interface-->Notifier
Firebase DB	To store data persistently	Palette-->Experimental-->Firebase DB

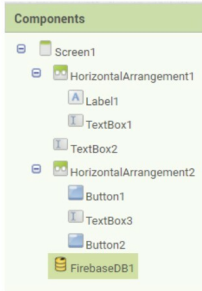


Figure 7: Components view

now ready. Figure 6 shows how the application will look after the installation.

8. The hierarchy of the components that we have dragged to the designer is shown in Figure 7. If you are confused seeing the designer and the components viewer, let me explain further. Here is the hierarchy that we have placed for our application.
 1. We have just placed four text boxes to get the details from the user.
 2. For easy reading, we have inserted a label before the text box.
 3. We need to configure the Firebase component to connect to the demo project we have created using the steps mentioned above. For that,

just assign the URL of your project to the Firebase URL property of the Firebase component in the designer.

Now let's head towards the blocks editor to define the various behaviours. Let's discuss the actual functionality that we are expecting from our application.

1. The user should be able to write a name in the top text box.
2. The user should be able to send the chat text by clicking on the *Send* button.
3. All connected users should see the chat history.
4. Upon clicking the button, chat history should be cleared.

So let's move on and add these features using the block editor. I hope you remember how to switch from designer to block editor. There is a button available right above the *Properties* pane to switch between these.

Block editor blocks

I have already prepared the blocks for you. All you need to do is drag the relevant blocks from the left side palette and drop them on the viewer. Arrange the blocks the same way you see them in the image. I will explain what each does and how it is called.

- On the first line, we have initialised two variables to get the user name and chat history.
- On clicking the *Clear history* button, all text from the *History* text box will be erased.

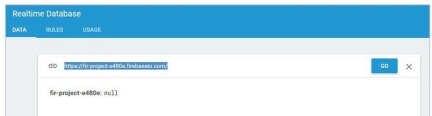


Figure 8: URL of the project